Predicting the Tommy Heinsohn Hustle Award

Model Comparison and Results

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1 Overview

This document presents the results of several classification models trained to predict the recipient of the Tommy Heinsohn Hustle Award on a per-game basis. We compare regularized logistic regression (ridge, lasso, elastic net), a decision tree, a random forest, and XGBoost on the held-out **2024–25** and **2025–26** seasons.

2 Sample game data

Table **??** lists one Celtics game from the hustle table (2026-03-06, game ID 22500908 in the source notebook export). The **Tommy winner** column is the per-game label `y` (1 = Tommy recipient for that game). **Net rtg.** is the on/off proxy `plusMinusPoints`, shown as a whole number as in the notebook.

Table 1: Sample box-style rows for 2026-03-06 (Boston).

Date	Player	Tommy win.	Min	PTS	REB	AST	OREB	STL	BLK	Net rtg.
2026-03-06	Payton Pritchard	0	33.8	18	7	6	2	0	2	19
2026-03-06	Jaylen Brown	0	33.3	24	7	7	3	2	0	20
2026-03-06	Derrick White	0	31.1	20	4	4	1	2	2	23
2026-03-06	Jayson Tatum	1	27.2	15	12	7	1	1	0	20
2026-03-06	Neemias Queta	0	25.6	16	15	0	3	1	1	16
2026-03-06	Sam Hauser	0	22.5	6	5	0	1	0	2	0
2026-03-06	Baylor Scheierman	0	20.8	5	2	2	0	0	0	0
2026-03-06	Hugo González	0	18.9	6	2	2	0	0	1	2
2026-03-06	Luka Garza	0	17.9	10	3	1	3	0	2	9
2026-03-06	Ron Harper Jr.	0	4.3	0	1	0	0	0	0	-4
2026-03-06	Amari Williams	0	2.7	0	0	0	0	0	0	-4
2026-03-06	Nikola Vučević	0	1.8	0	0	0	0	0	0	-1

3 Model Results

Table 1 compares every classifier on the held-out 2024–25 and 2025–26 seasons, sorted by Top-1 game accuracy from highest to lowest.

Table 2: All model results on the 2024–25 and 2025–26 test seasons, ranked by Top-1 game accuracy.

Model	**Top-1 Acc.**	**F1**	**Recall**	**PR-AUC**
Random forest	0.4233	0.4233	0.4233	0.3630
XGBoost	0.3988	0.3988	0.3988	0.3363
Ridge logistic	0.3926	0.3926	0.3926	0.3168
Elastic-net logistic	0.3804	0.3804	0.3804	0.3168
Lasso logistic	0.3497	0.3497	0.3497	0.3083
Decision tree	0.3067	0.3067	0.3067	0.2559

Table 2 presents a condensed view with only the Top-1 game accuracy for each model.

Table 3: Model Top-1 game accuracy on the 2024–25 and 2025–26 test seasons.

Model	**Top-1 Acc.**
Random forest	0.4233
XGBoost	0.3988
Ridge logistic	0.3926
Elastic-net logistic	0.3804
Lasso logistic	0.3497
Decision tree	0.3067

4 Random Forest Feature Importance

Table 3 lists the 15 most important features from the random-forest classifier (the best-performing model in Table 1).

Table 4: Top 15 feature importances from the random forest classifier.

Rank	Feature	Importance
1	points_per_min	0.0873
2	points	0.0750
3	fieldGoalsMade	0.0692
4	points_rank	0.0599
5	usage_rate	0.0454
6	fieldGoalsAttempted	0.0434
7	minutes_decimal	0.0391
8	reb_per_min	0.0349
9	threePointersMade	0.0345
10	impact_efficiency	0.0338
11	ast_per_min	0.0324
12	role_outperformance	0.0291
13	plusMinusPoints_rank	0.0273
14	hustle_proxy	0.0271
15	stocks_per_min	0.0261

5 Predicted Tommy Award Picks per Team

Table 4 lists each opposing team's most-predicted Tommy Award candidate, with the count of games for which the random-forest model picked that player as the team's top hustler in the 2024–25 and 2025–26 seasons.

Table 5: Top predicted hustler per team across the 2024–25 and 2025–26 seasons.

Team	**Player**	**2024–25**	**2025–26**	**Total**
Denver Nuggets	Nikola Jokić	50	50	100
Milwaukee Bucks	Giannis Antetokounmpo	49	29	78
San Antonio Spurs	Victor Wembanyama	29	41	70
Oklahoma City Thunder	Shai Gilgeous-Alexander	33	34	67
Detroit Pistons	Jalen Duren	22	38	60
New York Knicks	Karl-Anthony Towns	31	26	57
Houston Rockets	Alperen Sengun	31	23	54
Phoenix Suns	Devin Booker	29	22	51
Los Angeles Clippers	Ivica Zubac	39	11	50
Philadelphia 76ers	Tyrese Maxey	22	27	49
Los Angeles Lakers	Luka Dončić	10	38	48
New Orleans Pelicans	Zion Williamson	18	29	47
Golden State Warriors	Stephen Curry	26	17	43
Sacramento Kings	Domantas Sabonis	35	5	40
Minnesota Timberwolves	Rudy Gobert	19	21	40
Chicago Bulls	Nikola Vučević	25	14	39
Toronto Raptors	Scottie Barnes	14	25	39
Cleveland Cavaliers	Jarrett Allen	23	15	38
Memphis Grizzlies	Jaren Jackson Jr.	24	13	37
Indiana Pacers	Pascal Siakam	17	18	35
Portland Trail Blazers	Donovan Clingan	9	24	33
Miami Heat	Bam Adebayo	16	15	31
Orlando Magic	Paolo Banchero	15	15	30
Atlanta Hawks	Trae Young	27	3	30
Charlotte Hornets	LaMelo Ball	11	16	27
Utah Jazz	Lauri Markkanen	11	15	26
Washington Wizards	Alex Sarr	10	14	24
Dallas Mavericks	Daniel Gafford	15	8	23
Brooklyn Nets	Noah Clowney	7	10	17

6 Baseline Figures

Below are the baseline plots exported from the notebooks.

Tree baseline hyperparameters:

	criterion	max_depth	min_samples_split	min_samples_leaf	max_features	r
0	entropy	8	20	10	sqrt	

Decision tree metrics on test seasons:

	game_top1_accuracy	row_f1	row_recall	row_prauc	row_precision
0	0.239264	0.239264	0.239264	0.223978	0.239264

Figure 1: Baseline decision-tree results.

	n_estimators	max_depth	min_samples_split	min_samples_leaf	max_features
0	300	12	20	10	sqrt

Random forest metrics on test seasons:

	game_top1_accuracy	row_f1	row_recall	row_prauc	row_precision
0	0.417178	0.417178	0.417178	0.357337	0.417178

Top 15 feature importances:

	feature	importance
0	fieldGoalsMade	0.094992
1	points	0.089808
2	points_per_min	0.084961
3	points_rank	0.057328
4	fieldGoalsAttempted	0.041579
5	usage_rate	0.039936
6	minutes_decimal	0.036657
7	threePointersMade	0.036281
8	ast_per_min	0.032065
9	reb_per_min	0.031292
10	impact_efficiency	0.029285
11	role_outperformance	0.028572
12	stocks_per_min	0.027306
13	plusMinusPoints_rank	0.027106
14	hustle_proxy	0.026432

Figure 2: Baseline random-forest results.

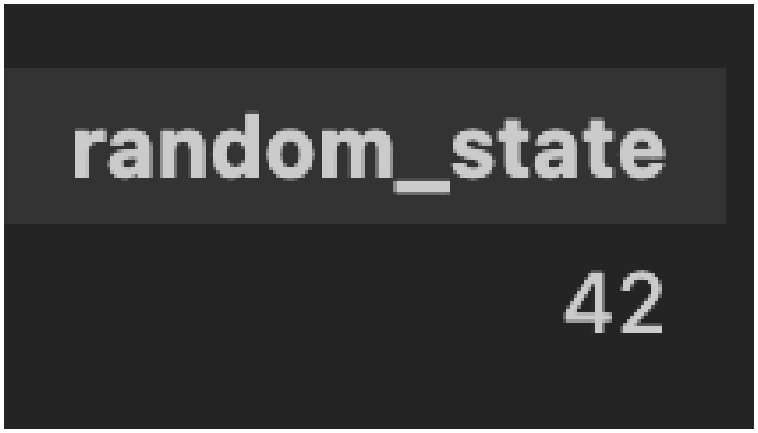

Figure 3: Decision-tree performance across random states.

7 Notes

- The `siunitx` package aligns the decimal points so numeric columns look clean; adjust `table-format` in the preamble if a column needs more/fewer digits.
- To add a new model's row, copy any existing row in Table 1 and replace the numbers.
- To add a new figure, drop the image file into the same folder as this `.tex` file and copy a `\begin{figure}...\end{figure}` block.